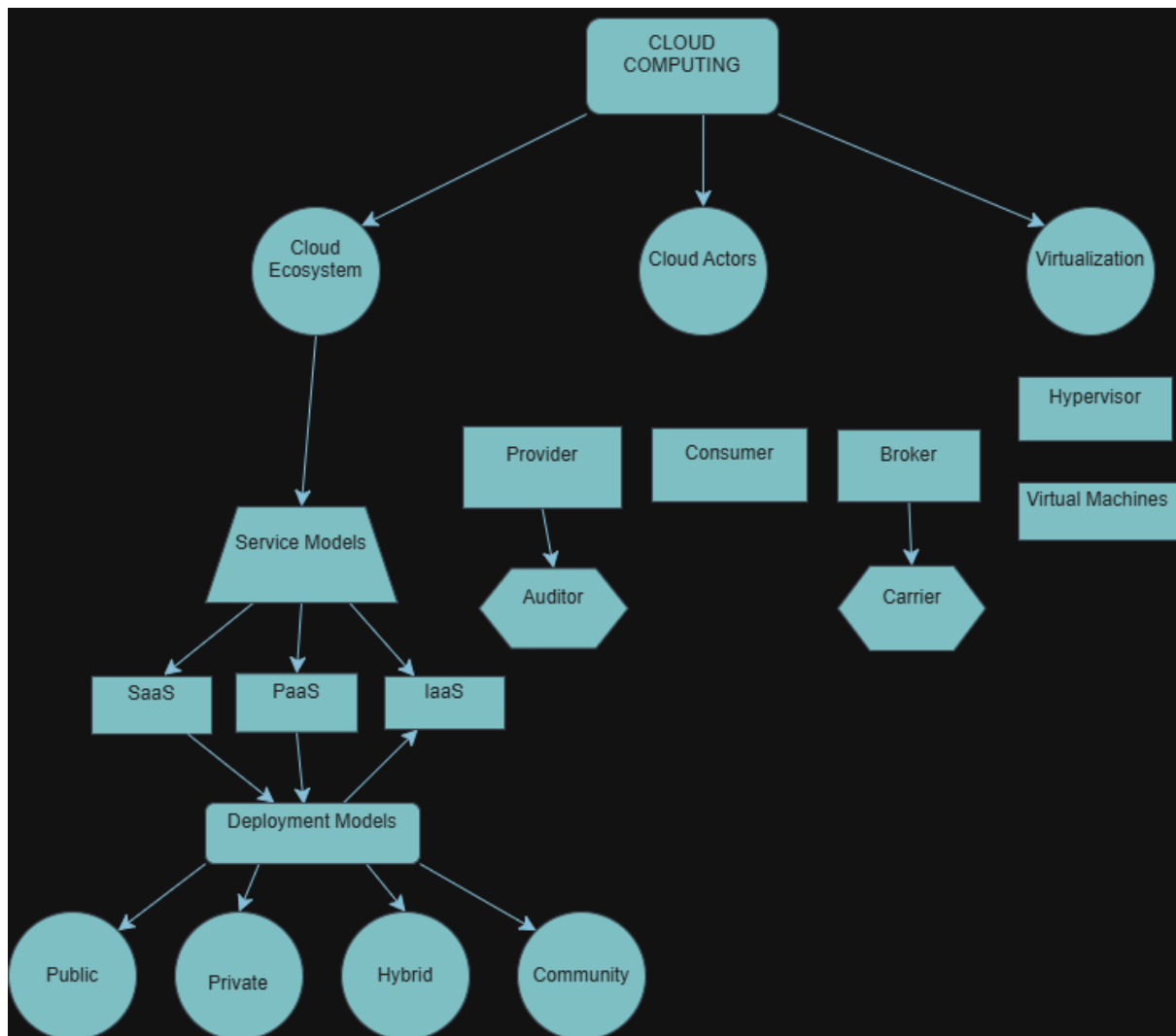


1.



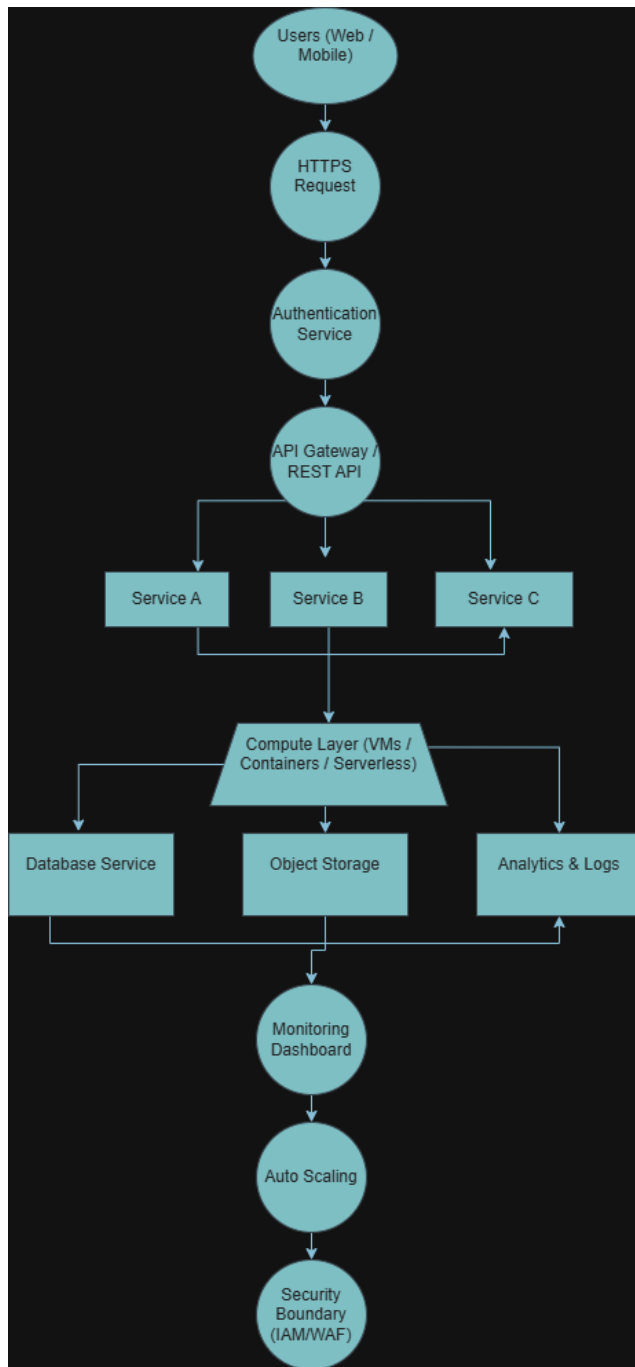
2A.

SaaS	Complete application, updates, maintenance	User configuration	Ready-to-use software
PaaS	Runtime, OS, middleware, scaling	Application code	Faster development
IaaS	Servers, storage, networking	OS, middleware, applications	Full infrastructure control
FaaS / Serverless	Infrastructure, scaling	Function code	Event-driven execution
Database as a Service	Database engine, backups, patching	Tables, queries, data	Easy database management
Storage as a Service	Storage hardware, replication	Files and objects	Secure file storage
Identity as a Service	Authentication platform	User roles and permissions	Secure access control
Monitoring & Logging	Metrics collection, dashboards	Alerts and log analysis	Performance monitoring

2B.

Project Module	Cloud Service	Purpose
Authentication	Identity as a Service	User login and authentication
Frontend	Static Website Hosting	Host web application
Backend/API	PaaS / Serverless	Business logic
Database	Database as a Service	Store application data
Object Storage	Storage as a Service	Store images and documents
Analytics & Logging	Monitoring Service	Collect logs and metrics
Notifications	Notification Service	Email/SMS alerts
AI/Python Module	Managed AI Service	Machine learning processing
Security	IAM + Firewall	Secure resources
CI/CD	DevOps Pipeline	Automated deployment
Backup	Backup Service	Disaster recovery
Monitoring	Monitoring Dashboard	Health monitoring

3.



4. Generic Technical Justification

The proposed cloud architecture demonstrates the effective implementation of core cloud computing concepts by integrating the cloud ecosystem, cloud actors, service models, deployment models, virtualization, web services, APIs, scalability, elasticity, and cloud security into a unified architecture. The concept map illustrates the relationships between these components and provides a clear understanding of how cloud technologies work together to support modern applications.

The cloud ecosystem consists of cloud service providers, consumers, brokers, auditors, and carriers, each performing a specific role in delivering, managing, and consuming cloud services. Understanding these cloud actors helps define responsibilities, service delivery, governance, and communication within a cloud environment. The deployment models—Public Cloud, Private Cloud, Hybrid Cloud, and Community Cloud—offer different levels of accessibility, security, cost, and control, allowing organizations to choose the most suitable environment based on their operational requirements.

The architecture utilizes different cloud service models to optimize resource utilization and simplify application management. **Software as a Service (SaaS)** delivers ready-to-use software without requiring infrastructure management. **Platform as a Service (PaaS)** provides a managed environment for developing and deploying applications without maintaining operating systems or middleware. **Infrastructure as a Service (IaaS)** offers virtual machines, networking, and storage, giving organizations complete control over their computing environment. **Function as a Service (FaaS) or Serverless Computing** enables event-driven execution where code runs only when triggered, reducing operational overhead and improving cost efficiency. Additional managed services such as Database-as-a-Service (DBaaS), Storage-as-a-Service (STaaS), Identity-as-a-Service (IDaaS), and Monitoring-as-a-Service further simplify application development and operations.

Virtualization forms the foundation of cloud computing by abstracting physical hardware into multiple virtual machines or containers using hypervisors. This enables efficient resource sharing, workload isolation, improved hardware utilization, rapid provisioning, and high availability. Managed compute services extend these benefits by allowing applications to run without direct server administration, enabling developers to focus primarily on application functionality.

Web services enable communication between distributed cloud components. REST (Representational State Transfer) APIs use lightweight HTTP methods and JSON data, making them suitable for web and mobile applications due to their simplicity, flexibility, and performance. SOAP (Simple Object Access Protocol) provides standardized XML-based messaging with enhanced security and reliability, making it suitable for enterprise applications requiring strict communication standards. APIs serve as the communication bridge between users, applications, cloud services, databases, storage systems, and external services, ensuring interoperability and seamless integration.

The architecture is designed with both **scalability** and **elasticity** as fundamental cloud characteristics. Scalability enables the system to handle increasing workloads by adding additional computing resources either **vertically** (increasing CPU, memory, or storage of existing servers) or **horizontally** (adding more server instances behind a load balancer). Elasticity automatically adjusts resource allocation according to real-time demand, increasing resources during peak traffic and releasing unused resources during periods of low activity. This dynamic resource management improves application performance while minimizing operational costs through the pay-as-you-use model.

The service responsibility matrices clearly illustrate the **Shared Responsibility Model**, where cloud providers manage the underlying infrastructure, networking, physical security, virtualization, hardware maintenance, software updates, backups, and platform availability, while the application team is responsible for application logic, business rules, data management, user access control, configurations, and application-level security. Clearly defining these responsibilities enhances system reliability, security, compliance, and operational efficiency.

The API flow architecture further demonstrates secure request processing, where users interact with the application through a web or mobile interface. Requests are authenticated using identity services before passing through an API Gateway, which securely routes traffic to backend services. These services communicate with managed databases, object storage, analytics, monitoring systems, and other cloud resources while operating within defined security boundaries. Monitoring and logging services continuously track application health, resource utilization, performance metrics, and security events, enabling proactive maintenance and automatic scaling.